1. Make this slide more relevant. Want to highlight the delay in detection and irreversible damage when diagnosis is normally done. Want to diagnose early and most are not as of right now. Dopamine once lost cant be fixed, cant be noticed until late stage. Look for brighter or better picture.
   a. Emphasize people are coming in for other things and tremor or other symptoms of parkinsons disease get noticed. This is too late.
   b. Show patient picture and picture of substantia nigra, what PD patient looks like.**
2. **Make sure to explain why we picked these two sources. What do each of them bring to the table. What do they offer specifically. How is it useful for us. What is the hypothesis.
   Why are we comparing these groups?
   Add in visuals
3. Use Flowchart** and then breakdown
   Make paired bar graph with PD positive: NAMCS/NHAMCS and PD Negative: NAMCS/NHAMCS
   Make sure everything is legible and large enough

   Change title to express the major reason of this slide. Not "results" but maybe "Major reason for visit"
4. Flow,future work, acknowledgement

https://www.dreamstime.com/parkinson-disease-symptoms-infographic-idea-dementia-neurology-illness-tremor-memory-loss-isolated-vector-illustration-image161160294
https://www.123rf.com/photo_94431365_stock-vector-parkinson-s-disease-cartoon-icon-vector-illustration-graphic-design-.html
https://depositphotos.com/176013252/stock-illustration-illustration-parkinson-disease-symptoms-sign.html
https://www.sciencesource.com/archive/Substantia-Nigra---Parkinson-s-Disease-SS2689192.html

Parkinson's Disease Patient Profiles in Different Healthcare Set-ups: A Comparative Analysis

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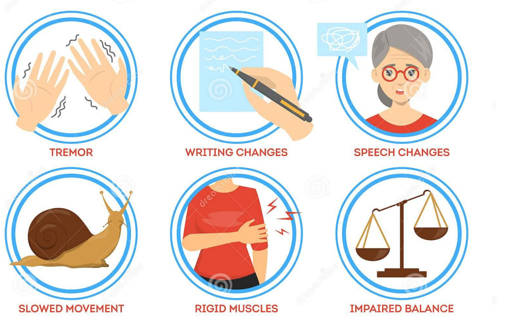
Parkinson’s Disease

- Second most common neurodegenerative disorder
- Predominantly motor manifestations: Affects posture and movement
- Caused by loss of dopamine producing neurons in the brain
- 50,000 new cases reported each year in the United States
- Disease progresses without clinical manifestations: Critical to diagnose in early stage



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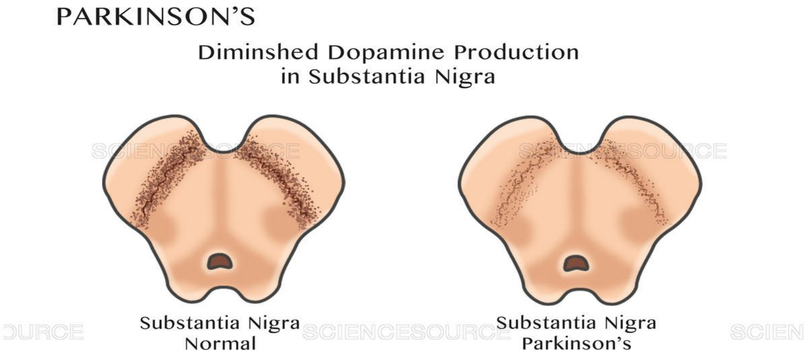

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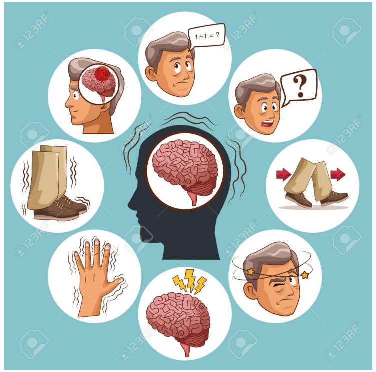
Purpose

To answer, how and when the disease is diagnosed in the US?

Comparative study:

- National Ambulatory Medical Care Survey (NAMCS)
- National Hospital Ambulatory Medical Care Survey (NHAMCS) -OPD



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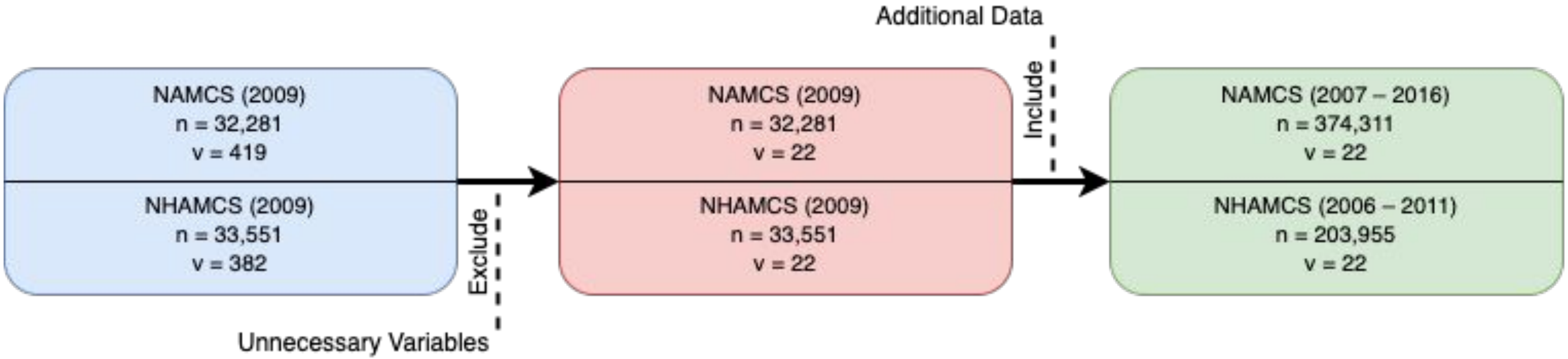
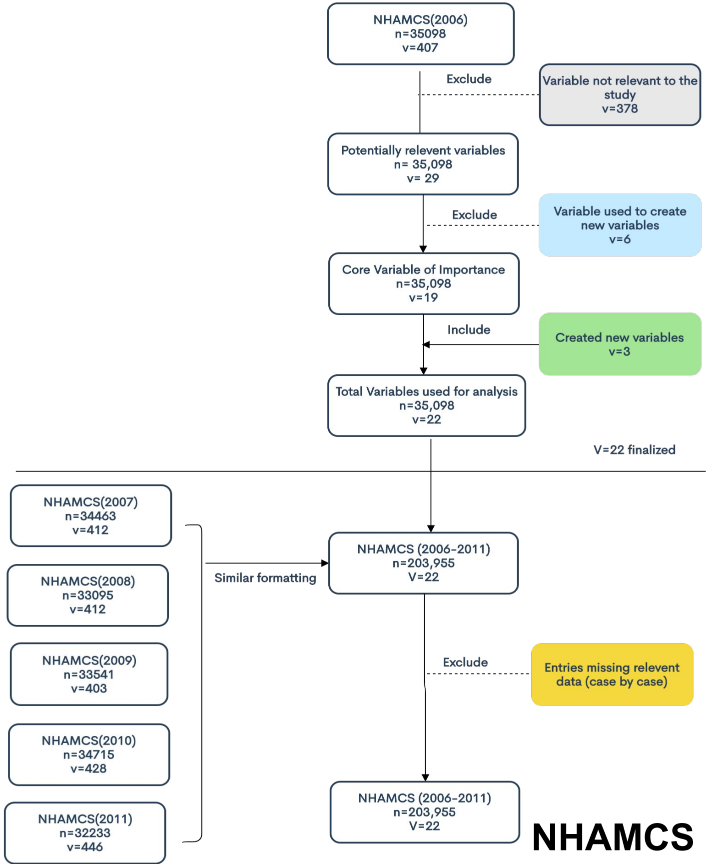
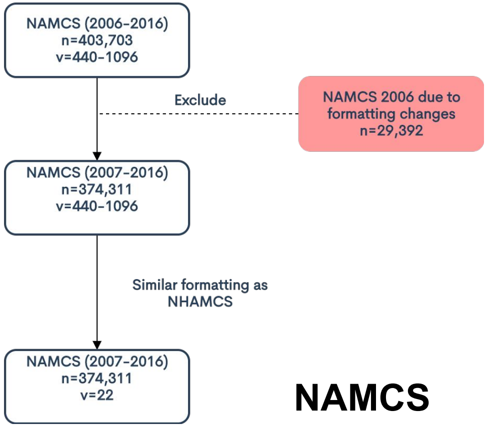
Data Preprocessing



Survey	PD patients	Total Sample Size
NAMCS	1036	374,311
NHAMCS	360	203,988

NAMCS and NHAMCS Data





Survey	PD patients	Total Sample Size
NAMCS	1036	374311
NHAMCS	360	203955

NAMCS and NHAMCS Data

- Sample Records

	Years	Sample size	Variables Per Year
NHAMCS	2006-2010	203,955	93-421
NAMCS	2007-2016	374,311	440-1,096

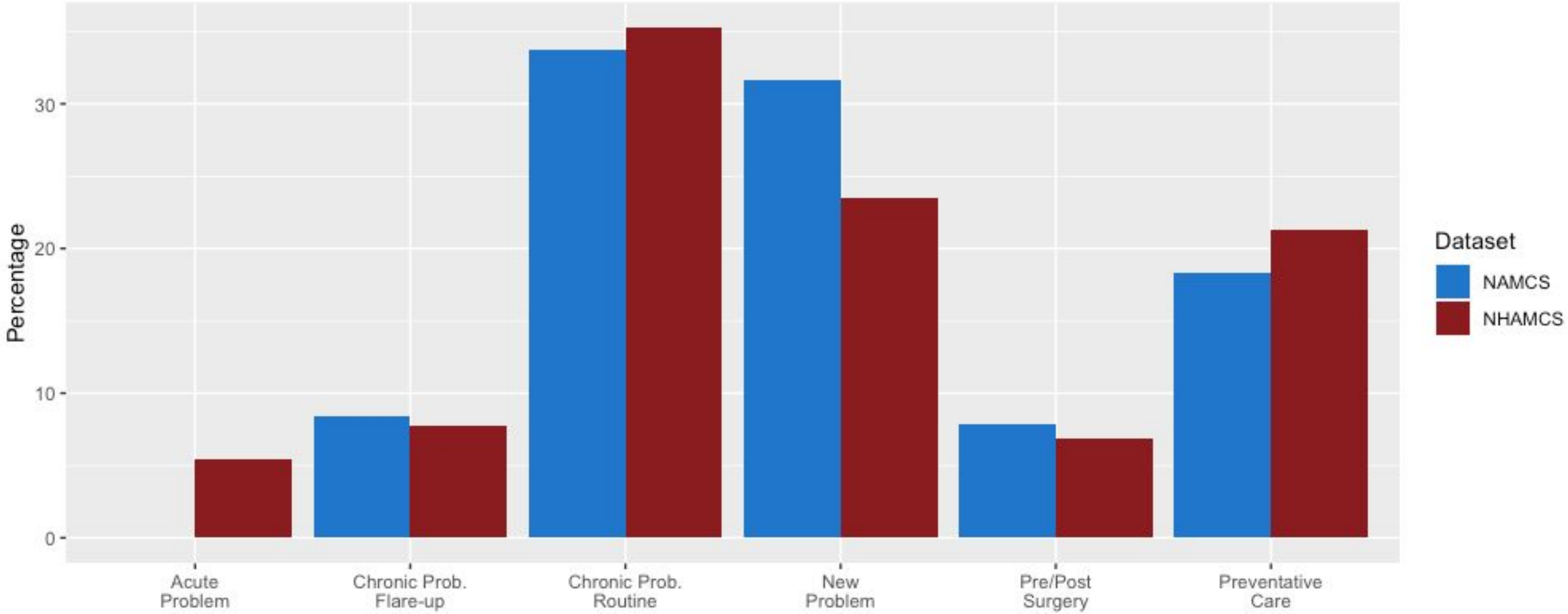
- Unweighted
- 23 variables
- The International Classification of Diseases (ICD-9-CM) was used to choose PD patient sample

Hidden Slide

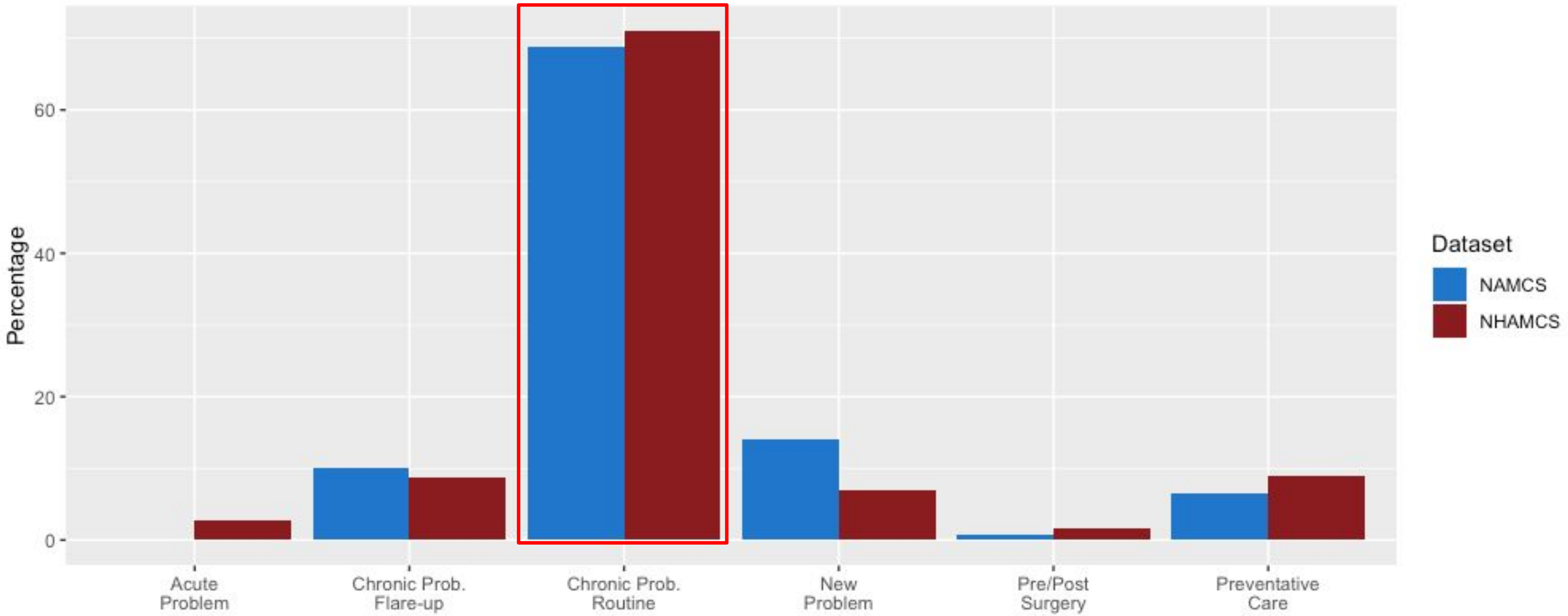
Year	NAMCS			NHAMCS		
	PD positive	PD Positive(+RFV)	Sample Size	PD positive	PD Positive(+RFV)	Sample Size
2006				53	57	35098
2007	154	166	32778	42	46	34463
2008	92	97	28741	43	46	33905
2009	95	99	32281	50	53	33541
2010	117	123	31229	77	80	34715
2011	120	124	30872	76	78	32233
2012	112	121	76330	341	360	203955
2013	113	122	54873			
2014	61	70	45710			
2015	83	85	28332			
2016	0	29	13165			
	947	1036	374311			

Lets put this if itis necessary.

Major Reason for Visit: PD Negative Population

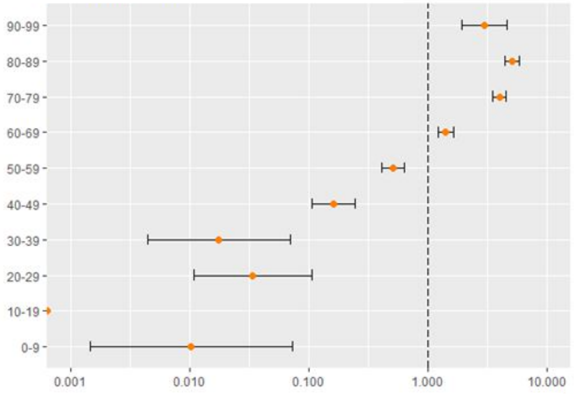


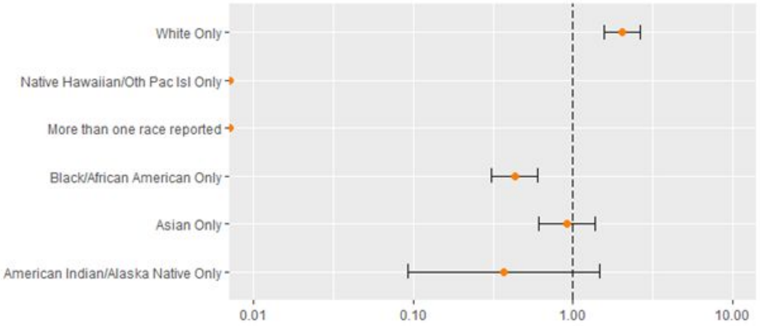
Major Reason for Visit: PD Positive Population



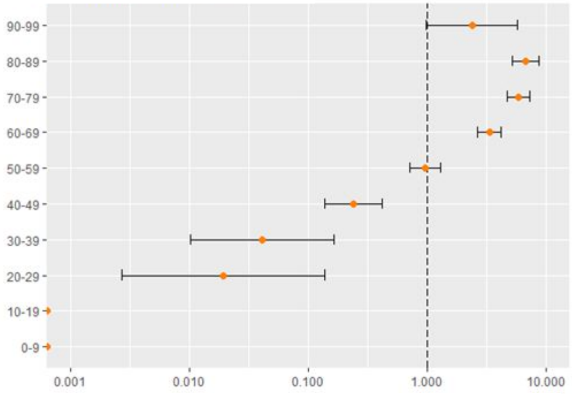
Comparison of Demographics

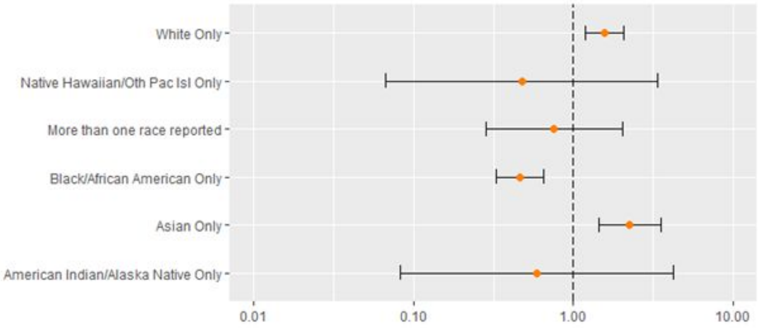
NAMCS





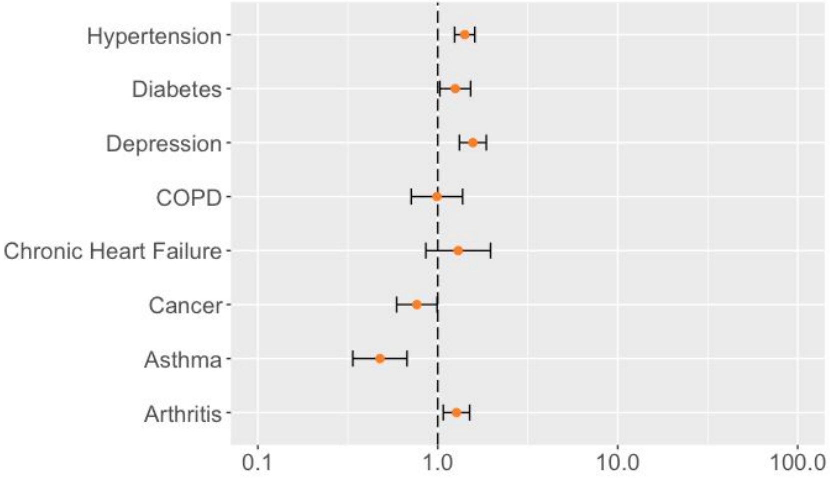
NHAMCS

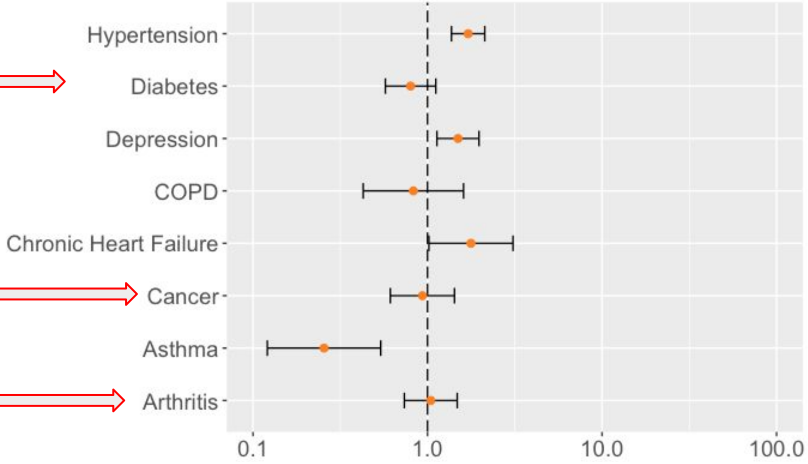


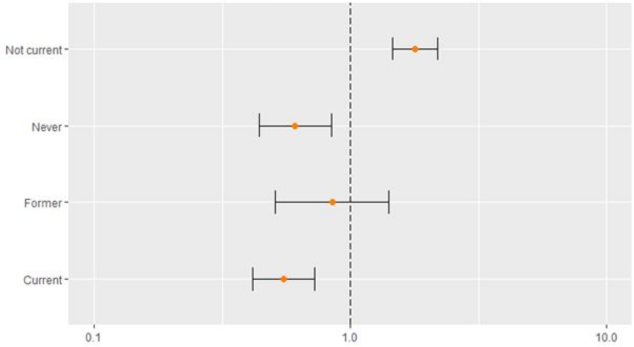


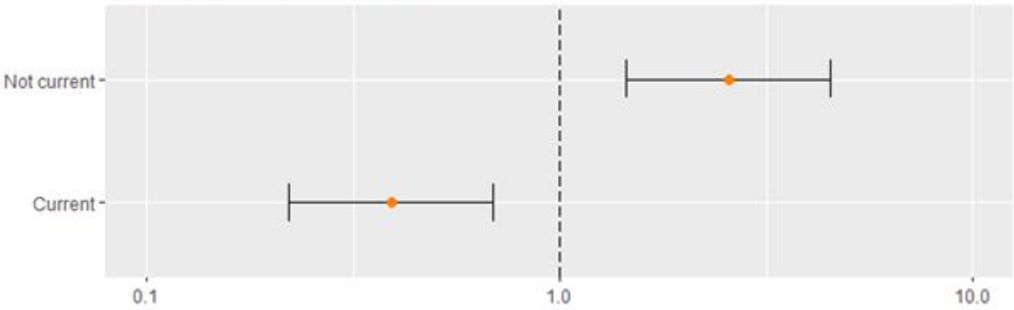
Comparison of Comorbidities

NAMCS NHAMCS









Discussion

Similarities	Differences
● Strong correlation with PD: ○ Male ○ Caucasian ○ 60-99 year olds ○ Hypertension ○ Depression ● Inverse relationship with PD: ○ Smoking ○ Asthma ● No relationship with PD: ○ COPD	● Comorbidities: ○ Arthritis ○ Diabetes ○ Cancer

Future Work

- Separate those with PD into above and below 60 years old
    - Search for commonalities unique to the younger population

- Build a predictive model using variables with similar relationships amongst both datasets

Acknowledgements

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  Prof. Sumona Mondal, Department of Mathematics
  Prof. Shantanu Sur, Department of Biology

- Colleagues:
  Daniel Fuller

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